

Mohamed Khalil ZRELLY

Generative AI Engineer Apprentice | Computer Science Engineering Student

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Professional Summary

Generative AI Engineer Apprentice at Crédit Agricole, specializing in agentic workflow orchestration. Computer Science Engineering student at Sup Galilée, Sorbonne Paris North University, with hands-on expertise in RAG, NLP, and reinforcement learning.

Work Experience

Crédit Agricole Personal Finance & Mobility

Generative AI Engineer Apprentice

France

09/2025 — Present · 10 months

- Developed a declarative, config-driven microservice for **workflow management and orchestration**, making it easy to manage and execute complex AI workflows at scale.
- Built an automated **workflow evaluation engine** leveraging **LLM-as-a-judge** to assess outputs against custom metrics, with a **dashboard** to analyse workflow behaviour.

Agentic Workflows · Advanced RAG · Amazon Bedrock · LLM Evaluation · Observability · Python/FastAPI · Amazon DynamoDB · Python/Streamlit

VERMEG for Banking & Insurance Software

Natural Language Processing Intern

Tunisia

02/2024 — 06/2024 · 5 months

- Implemented a system that automates email classification and information extraction from attachments, using fine-tuned **BERT** and **OCR** to improve accuracy and optimise efficiency.
- Developed a web application for **BERT Fine-Tuning**, email management, and real-time data analysis and visualisation, enabling model experimentation and performance monitoring.

Hugging Face Transformers · Sentence-Transformers · Fine-Tuning · PyTorch · PaddleOCR · Python/Flask · MongoDB · React/Redux · Bootstrap

Sofiatech

Machine Learning Intern

Tunisia

06/2023 — 08/2023 · 3 months

- Evaluated multiple **reinforcement learning** algorithms from **Stable Baselines 3** across various **Gymnasium** environments to assess agent performance and training efficiency.
- Built a **2D game in Pygame** where a **Deep Q-Learning** agent learns to navigate a dynamic environment, applying the best-performing algorithm from the benchmarking phase.

Reinforcement Learning · Deep Q-Learning · stable-baselines3 · TensorFlow/Keras · Gymnasium · NumPy · Pandas · Matplotlib · Python/Pygame

Featured Projects

CVision | RAG · CV Recommendation System

In Production — 2025

Created an intelligent **CV recommendation system** based on job description using **Retrieval-Augmented Generation (RAG)**, to optimise the selection and automatic ranking of the most relevant profiles.

Advanced RAG · Re-Ranking · LangChain · Qwen3-VL · Milvus · Python/Streamlit · Sentence-Transformers · PDF Parsing · Prompt Engineering

Twixel | Multi-Agent · Social Listening System

In Ideation · 2026

Designing an intelligent **multi-agent system** for automated social listening on Twitter (now X), aimed at identifying emerging trends in real time and generating actionable marketing campaign ideas at scale.

Multi-Agent Systems · LangGraph · MCP/A2A · GPT-OSS · Qdrant · Python/FastAPI · Vue.js · X API · Trend Detection · Docker · Social Listening

Technical Skills

Generative AI & Agentic Systems	LangChain, LangGraph, Milvus, Qdrant, Amazon Bedrock
LLM Observability & Evaluation	OpenTelemetry, Prometheus, Grafana, DeepEval, Langfuse
NLP, LLM Fine-Tuning & RLHF	Transformers, Datasets, PEFT, Accelerate, TRL, Spacy
Machine Learning, DL & MLOps	Sklearn, PyTorch, TensorFlow, MLflow, Amazon SageMaker
Full-Stack Web Development	Vue.js, FastAPI, Flask, React, Express.js, NestJS
Data Engineering & Databases	PySpark, Airflow, PostgreSQL, MongoDB, Amazon DynamoDB
Cloud Deployment & CI/CD	Git, Docker, GitHub Actions, Terraform, Amazon ECR/ECS
Programming Languages (PL)	Python, JavaScript/TS, SQL, C/C++, Rust, Java, PHP

Education

Sup Galilée – Sorbonne Paris North University

Engineering Degree in Computer Science

Villetaneuse, France

2024 — 2027 (Expected)

ISI Ariana – University of Tunis El Manar

Bachelor's Degree in Computer Science

Ariana, Tunisia

2021 — 2024 · High Distinction

Certifications & Languages

IBM (Coursera)	Build Multimodal Generative AI Applications · Generative AI Advanced Fine-Tuning for LLMs
Udemy	Spark and Python for Big Data with PySpark · Natural Language Processing with Python
Microsoft Azure	Azure AI Fundamentals (AI-900) · Azure Data Fundamentals (DP-900)
Spoken Languages	English — B2 · TOEIC French — B2 · TCF Arabic — Native